

Please ensure this tracking sheet is completed as a group, as it will be used to monitor your individual participation within the team.

Team Members	Role/ Responsibility	Email				
Mildred Ewenrim Ebomah	Team Lead	m.ebomah@alustudent.com				
Ange Gabriella Ahirwe		g.ahirwe@alustudent.com				
Sheja Queen Linda		l.sheja@alustudent.com				
Ntagungira Rachid						

Task Allocation and Tracker

	Task Allocated	Assigned Member	Deadline	Completion Status (Not Started/In Progress/Completed/Did not participate)	Reviewed by Team? (Yes/No)	Comments
	Part 1 --Probability Distributions	Mildred	29/06/26	Completed	Yes	
	Part 2 --Bayesian Probability	Gabriela	30/06/26	Completed	Yes	Choose 2-4 positive and 2-4 negative keywords from the IMDb reviews dataset; compute Prior, Likelihood, Marginal and Posterior for each keyword using Bayes' Theorem implemented in plain Python (no ML libraries).
	Part 3 --Gradient Descent Manual Calculation	Linda	01/07/26	Completed	Yes	Manually compute predicted y , derive MSE gradients w.r.t. m and b using matrix multiplication, and perform gradient descent updates (one per group member), showing all chain-rule steps in a neat PDF.
	Part 4 --Gradient Descent in Code	Rachid	01/07/26	Completed	Yes	Convert manual gradient descent into Python code using SciPy for derivatives; update m and b , compute final predictions, and plot parameter/error changes over iterations with Matplotlib.

Meeting Attendance Log

Meeting Date	Agenda	Facilitator	Attendees	Key Discussion Points	Next Steps	
27/06/26	Assignment Breakdown	Mildred Ewenrim Ebomah	All team members	1. Each person works on different parts of the assignment. Part 1 --Mildred Part 2 --Gabby Part 3 -- Linda Part 4 --Ali 2. Ali is to create a repo and add everyone as contributors 3. Work on your jupitar notebook and we will merge 4. We have a check-in meeting on Monday by 8pm	1. Ali is to create a repo and add everyone as contributors 2. We have a check-in meeting on Monday by 8pm	
29/06/26	Progress Check-in	Ange Gabriella Ahirwe	All team members	Each member gave a progress update. Gabriela shared her shortlisted positive/negative keywords for the Bayesian analysis; group flagged that one keyword appeared too rarely in the dataset and agreed to swap it. Linda walked through her first manual gradient descent update; a sign error in the $\partial J / \partial m$ derivation was spotted and corrected. Rachid showed early SciPy derivative code and asked for help debugging a matrix shape mismatch, which Mildred helped resolve.	Gabriela to finalize keyword list and compute posterior probabilities; Linda to redo iteration 2 and 3 with corrected gradient; Rachid to fix matrix dimensions and add plotting code; Mildred to continue EM iterations and prep tracking table.	

7/1/2026	Cross-Review & Corrections	Sheja Queen Linda	All team members	Team cross-reviewed each other's parts. Found the posterior probability formula in Part 2 was missing the marginal P (keyword) term and corrected it together. Part 4 plots were not updating per iteration due to a loop bug, fixed live on a call. Part 3 handwritten PDF was reviewed for legibility and chain-rule visibility; two steps were re-written more clearly. Part 1 tracking table values were checked against the printed code output and matched.	Rachid to push the fixed plotting code to the shared repo; Linda to rescan and re-upload the corrected PDF; everyone to review the full notebook before merging; Mildred to consolidate all parts into one notebook.	
7/3/2026	Final Rehearsal & Repo Merge	Ntagungira Rachid	All team members	Merged all four parts into the shared team repository and confirmed everyone appears in the git commit history. Did a full run-through of the 15-minute presentation and timed each section; Part 2 was running long so Gabriela trimmed her explanation. Assigned exact speaking order and Q&A responsibility per part. Confirmed README and contributions PDF were complete.	Book the presentation slot and add all members to the calendar invite; do one final rehearsal the day before; Rachid to do a last check that the repo runs end-to-end with no errors.	